



PROJECT SYNOPSIS

Project Title: AI-Based Organ Donor and Recipient Matching System Using AI and Machine Learning

Name: Rajesh Reddy

Registration Number: P19MT24S126082

Guide: Mr. Gowtham Kumar

Department: Master of Computer Applications (MCA)

Introduction

The AI-Based Organ Donor and Recipient Matching System is an intelligent healthcare platform designed to improve the organ transplantation process using Artificial Intelligence (AI) and Machine Learning (ML). Organ transplantation is one of the most effective treatments for patients suffering from organ failure. However, finding a suitable donor is often a challenging and time-consuming process due to compatibility requirements and the limited availability of organs. Traditional matching methods involve manual analysis and may delay life-saving transplants. The proposed AI-Based Organ Donor and Recipient Matching System uses Artificial Intelligence and Machine Learning to analyze donor and recipient data, identify compatible matches, and provide intelligent recommendations. This system helps improve the efficiency, accuracy, and transparency of the organ allocation process.

Problem Statement

Traditional organ matching methods rely on manual verification and limited matching criteria, which can delay transplantation and reduce matching efficiency. There is a need for an intelligent system that can automatically analyze donor and recipient information to provide accurate and timely organ matching.

Objectives

- Match organ donors with compatible recipients using AI and ML.
- Analyze donor and recipient medical information.
- Improve the accuracy of organ matching.
- Reduce the waiting time for transplantation.
- Generate compatibility scores and matching reports.
- Support healthcare professionals in organ allocation.

Scope

The system provides a centralized platform for managing donor and recipient information. It analyzes factors such as blood group, organ type, HLA compatibility, age, medical history, and location to recommend the most suitable recipient. The platform can be used by hospitals, transplant centers, blood banks, and healthcare organizations to improve the efficiency and transparency of organ transplantation.

Tools / Technologies

- **Programming Language:** Python
- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Backend:** Streamlit
- **AI & ML:** TensorFlow, Scikit-learn, Pandas, NumPy
- **Database:** SQLite

Hardware Requirements

- **Processor:** Intel Core i3 or above
- **RAM:** 8 GB (Minimum)
- **Storage:** 20 GB Free Disk Space
- **Internet Connection:** Required for accessing the system and database
- **Operating System:** Windows 10/11 (64-bit)

Expected Outcome

- Match organ donors and recipients with high accuracy.
- Reduce organ matching time.
- Improve transplantation success through AI-based recommendations.
- Generate compatibility scores and matching reports.

- Support transparent and efficient organ allocation.
- Improve healthcare services and patient outcomes.

Conclusion

The **AI-Based Organ Donor and Recipient Matching System** provides an intelligent and efficient approach to organ transplantation by using Artificial Intelligence and Machine Learning for donor-recipient matching. The system helps identify compatible recipients accurately, reduces the time required for organ allocation, and supports healthcare professionals in making informed decisions. By improving the speed, transparency, and reliability of the matching process, the proposed system has the potential to enhance transplant success rates and contribute to better patient care.